

**SULIT**

**DP024**

*Physics 2*

*Semester 2*

*Session 2023/2024*

*2 hours*

**DP024**

*Fizik 2*

*Semester 2*

*Sesi 2023/2024*

*2 Jam*



KEMENTERIAN PENDIDIKAN

KOLEJ MATRIKULASI KEDAH

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**PRA PEPERIKSAAN SEMESTER PROGRAM MATRIKULSI**

*MATRICULATION PROGRAMME EXAMINATION*

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**JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU**

*DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO*

**SET 3**

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Kertas soalan ini mengandungi 7 halaman bercetak

*This question paper consists of 7 printed pages*

- 1 (a) A progressive wave equation is given by  $y(x,t) = 0.6 \sin(8\pi t + 0.4x)$  where  $y$  and  $x$  are in cm and  $t$  in s. Calculate the following from this wave equation:

- i) Amplitude of the wave.
- ii) Direction of the wave propagation.
- iii) Frequency of the wave.
- iv) Period.
- v) Wave number.
- vi) Wavelength.
- vii) Velocity of particle when  $t = 0.3$  s and  $x = 0.3$  cm.

[11 marks]

- (b) Two waves,  $y_1$  and  $y_2$  propagating in opposite direction are given by the equations  $y_1 = 20 \sin(15t - 2x)$  and  $y_2 = 20 \sin(15t + 2x)$  where  $y$  and  $x$  are in meters and  $t$  in second. These two waves undergo superposition.

- (i) Name the resultant wave.
- (ii) Determine the equation of the resultant wave.

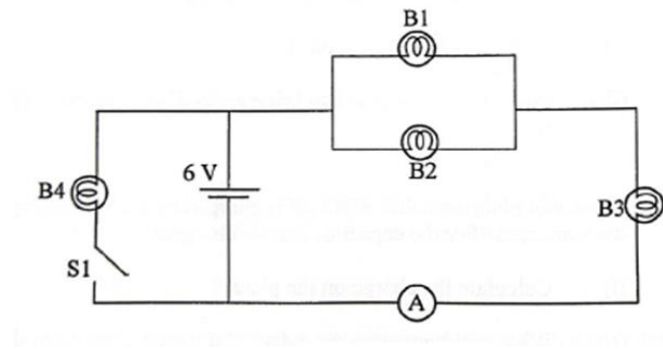
[2 marks]

- (c) A string that is exactly two meters long has a mass of 10.0 g. A transverse wave travels along this string at a speed of  $50.0 \text{ m s}^{-1}$ . Determine

- (i) The mass per unit length of the string..
- (ii) The tension of the string.

[3 marks]

- 2 (a) How many electron per second pass through a particular section of wire carrying a curent 0.50 A? [2marks]
- (b) You are given two resistors of 1 k $\Omega$ . How do you connect the resistors to a circuit that requires a 500  $\Omega$  resistance? Shows your suggestion [3 marks]
- (c)



**FIGURE 2**

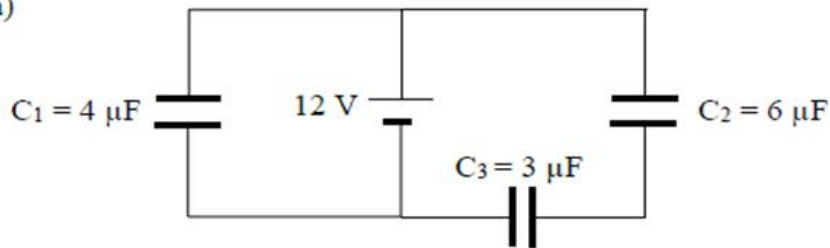
**FIGURE 2** shows four identical bulbs connected to a 6 V battery and a switch. When the switch is off, the ammeter reading is 0.5 A.

- (i) Calculate the resistance of a bulb.
- (ii) What happen to the reading of the ammeter when the switch in on? Explain your answer.

[ 7 marks]

3.

(a)



**FIGURE 4**

**FIGURE 4** shows three capacitor connected to a 12 V battery.

- (i) Determine the total capacitance of the circuit.
- (ii) Calculate the charge stored in  $C_3$ .

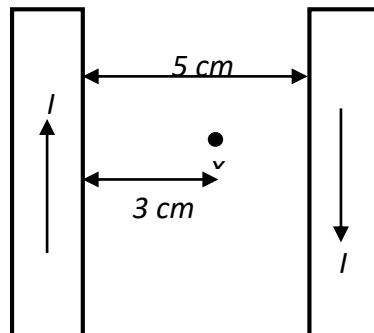
[4 marks]

- (b) A fully charged capacitor has  $12.0 \mu\text{F}$  and  $6.0 \text{ mC}$  charge. The capacitor has been discharged through  $100 \Omega$  resistors. Determine

- (i) the potential difference across the capacitor.
- (ii) the time constant for the discharging circuit and
- (iii) time taken when only 20% charges left in the capacitor.
- (iv) Sketch  $Q-t$  to show the discharging process.

[6 marks]

4. a) Two identical straight wire is carrying a current of  $I_1 = 3.0 \text{ mA}$  and  $I_2 = 5.0 \text{ mA}$  in the upwards direction as shown in the **Figure 1**.



**Figure 1**

Calculate the magnitude of the magnetic field at point  $x$

[4 marks]

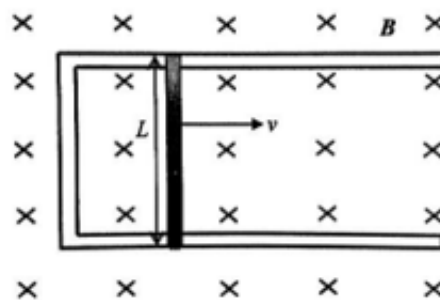
- b) A  $0.3 \text{ T}$  magnetic field is directed parallel to the plane of a circular loop of radius  $0.4 \text{ m}$ . Calculate the magnetic flux through the loop.

[3 marks]

- c) A circular coil of 180 turns and area  $2.5 \text{ cm}^2$  is placed in a changing magnetic field. The rate of change of magnetic field is  $2.60 \text{ T s}^{-1}$ . Calculate the induced emf in the coil.

[2 marks]

- d)



**Figure 2**

**Figure 2** shows a conductor of length  $l = 0.075 \text{ m}$  moves perpendicular in a uniform magnetic field  $2.0 \text{ T}$ . The emf induced in the conductor is  $0.46 \text{ V}$ .

- Calculate the speed of the conductor.
- If the total circuit resistance is  $0.6 \Omega$ , determine the magnitude and direction of the induced current.

[5 marks]

5. a) An object 4.0 cm high is placed 15.0 cm from a convex mirror of focal length 5.0 cm.

Determine

- (i) The position of the image from the mirror.
- (ii) The magnification of the image.
- (iii) The height of the image.
- (iv) Sketch the ray diagram to show the formation of the image

[9 marks]

- b) An object of height 10 cm is placed at 30 cm from a lens. The height of the virtual image formed is 20 cm. Calculate

- (i) the image distance.
- (ii) the focal length of the lens.
- (iii) state the type of lens

[5 marks]

6. (a) In a Young's double slit experiment, the separation between the double slits is 0.15 mm. A monochromatic light of wavelength 600 nm is used and the interference pattern is observed on a screen kept 1.5 m away.
- (i) Calculate the distance between central bright to the fourth order maximum.
  - (ii) Calculate the distance between third order minimum and the fifth order minimum.
- [7 marks]
- (b) In Young's double slit experiment using light of wavelength  $\lambda_1 = 480$  nm, interference fringes of 3 dark fringe width  $y_m$  are obtained. When the wavelength of light is increased to  $\lambda_2 = 600$  nm, the interference fringes of 5 dark fringe obtained and to overlap with 3 dark fringe. Find the ratio of the distance between slit to the screen ( $D_1 : D_2$ ) of the interfering sources in the two arrangements.
- [3 marks]
- (c) A 589 nm sodium light illuminates a double-slit with slit separation 0.4 mm. The screen is placed 1.5 m from the slits.
- (i) Calculate the distance between two successive bright fringes on the screen.
  - (ii) If the light with longer wavelength is used in this experiment, state what do you expect to observe. Give reason.
- [4 marks]